Project Interim Report

On

Hotel Room Reservation System Application

(Using Core Java & OOP Concepts)

RU-100-01-00012

Object Oriented Programming with Java

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Abstract

Hotel Room Reservation System is an Object-Oriented Programming based application designed to manage room bookings efficiently. The system allows users to book rooms, cancel reservations, check room availability, and calculate the total bill based on the number of days and room type.

This project is needed to reduce manual errors and simplify hotel management processes. It replaces traditional booking methods with a structured and automated approach.

The application uses core OOP concepts such as classes, objects, encapsulation, interfaces, and arrays. Rooms are stored using a two-dimensional array to represent floors and room numbers. The system ensures proper tracking of room status and provides accurate billing functionality.

Introduction

The Hotel Room Reservation System is designed to automate the process of room booking in a hotel. In traditional systems, booking and record management are done manually, which is time-consuming and error-prone.

This system helps:

Manage room records
Track room availability
Handle booking and cancellation
Calculate billing

Objectives

- To develop a system for managing hotel room reservations efficiently
- To implement booking and cancellation of rooms
- To track room availability in real-time
- To calculate the total bill based on duration of stay
- To reduce manual work and minimise errors
- To apply Object-Oriented Programming concepts such as classes, objects, and interfaces
- To design a structured and user-friendly system

Features of the Project

- Room booking functionality
- Cancellation of booked rooms
- Real-time room availability tracking
- Billing system based on number of days and room rate
- Organized storage of rooms using a 2D array
- Simple and user-friendly system design
- Error handling for already booked rooms
- Efficient management of multiple rooms and floors

Advantages of the Project

- Reduces manual work in hotel management
- Minimizes human errors in booking and billing
- Provides quick and efficient room reservation
- Easy to use and understand system
- Ensures accurate tracking of room availability

- Saves time in managing hotel operations
- Uses structure and organized approach with OOP concepts
- Can be easily upgraded with advanced features

Scope

- Suitable for small to medium-sized hotels
- Manages room booking and cancellation efficiently
- Tracks room availability in real-time
- Calculates total bill based on number of days and room type
- Reduces manual work and human errors
- Provides organized and systematic hotel management

Future Scope:

- Can be extended with database integration
- Online booking system can be added
- User login and authentication features
- GUI (Graphical User Interface) for better usability
- Payment gateway integration

System Requirements

Hardware:

- Computer/Laptop
- Minimum 4 GB RAM

Software:

- Operating System (Windows/macOS/Linux)
- Java JDK
- IDE (VS Code / Eclipse)

Methodology

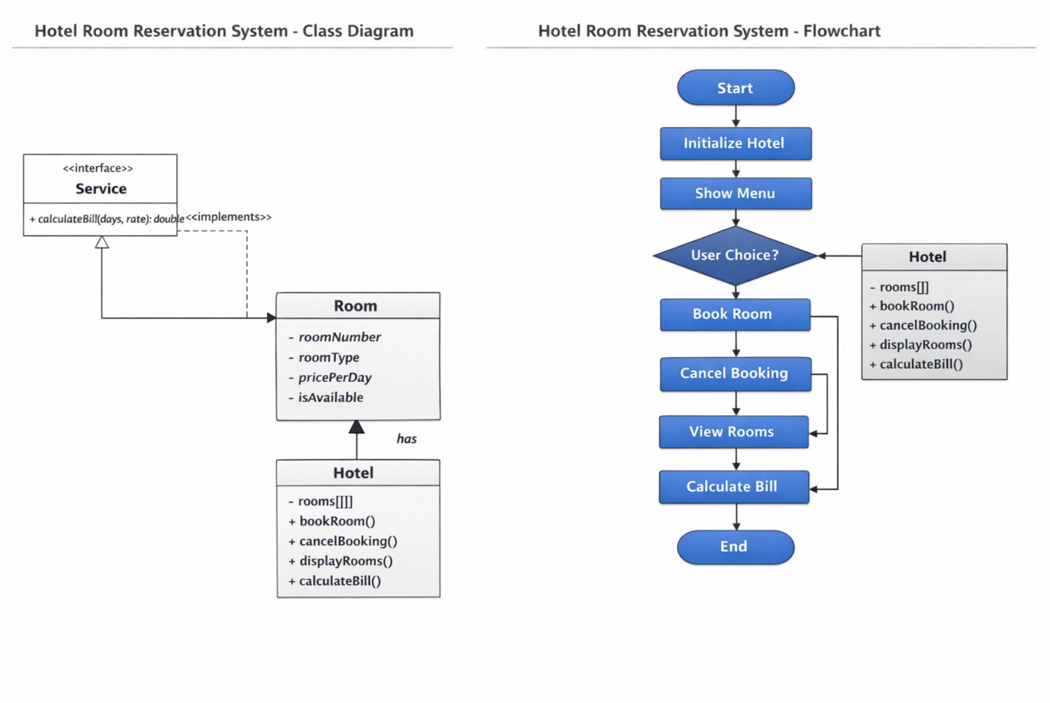
The Hotel Room Reservation System follows a structured approach to manage room bookings efficiently. Initially, the system is set up by creating rooms using a two-dimensional array to represent floors and room numbers.

The system then displays available rooms to the user and provides options such as booking a room, canceling a reservation, viewing room status, and calculating the total bill. Before booking, the system checks the availability of the selected room. If available, the room is booked and its status is updated; otherwise, an appropriate message is shown.

Similarly, when a booking is canceled, the room is marked as available again. The total bill is calculated based on the number of days and the room rate.

This step-by-step process ensures smooth and efficient hotel management.

Flowchart



Flowchart of Hotel Room Reservation System

Implementation

The Hotel Room Reservation System is implemented using Object-Oriented Programming concepts and consists of three main components: the Room class, the Hotel class, and the Service interface.

The **Room class** is used to store the details of each room, such as room number, room type, price per day, and availability status. Each room object represents a single room in the hotel.

The **Hotel class** is responsible for managing all the rooms. It uses a two-dimensional array to represent multiple floors and rooms. This class performs all major operations such as booking a room, canceling a reservation, displaying room status, and managing room availability. When a user tries to book a room, the system first checks whether the room is available. If available, it is marked as occupied; otherwise, an appropriate message is displayed. Similarly, when a booking is canceled, the room is marked as available again.

The **Service interface** defines the billing functionality. It contains a method to calculate the total bill, which is implemented in the Hotel class. The bill is calculated based on the number of days and the room rate.

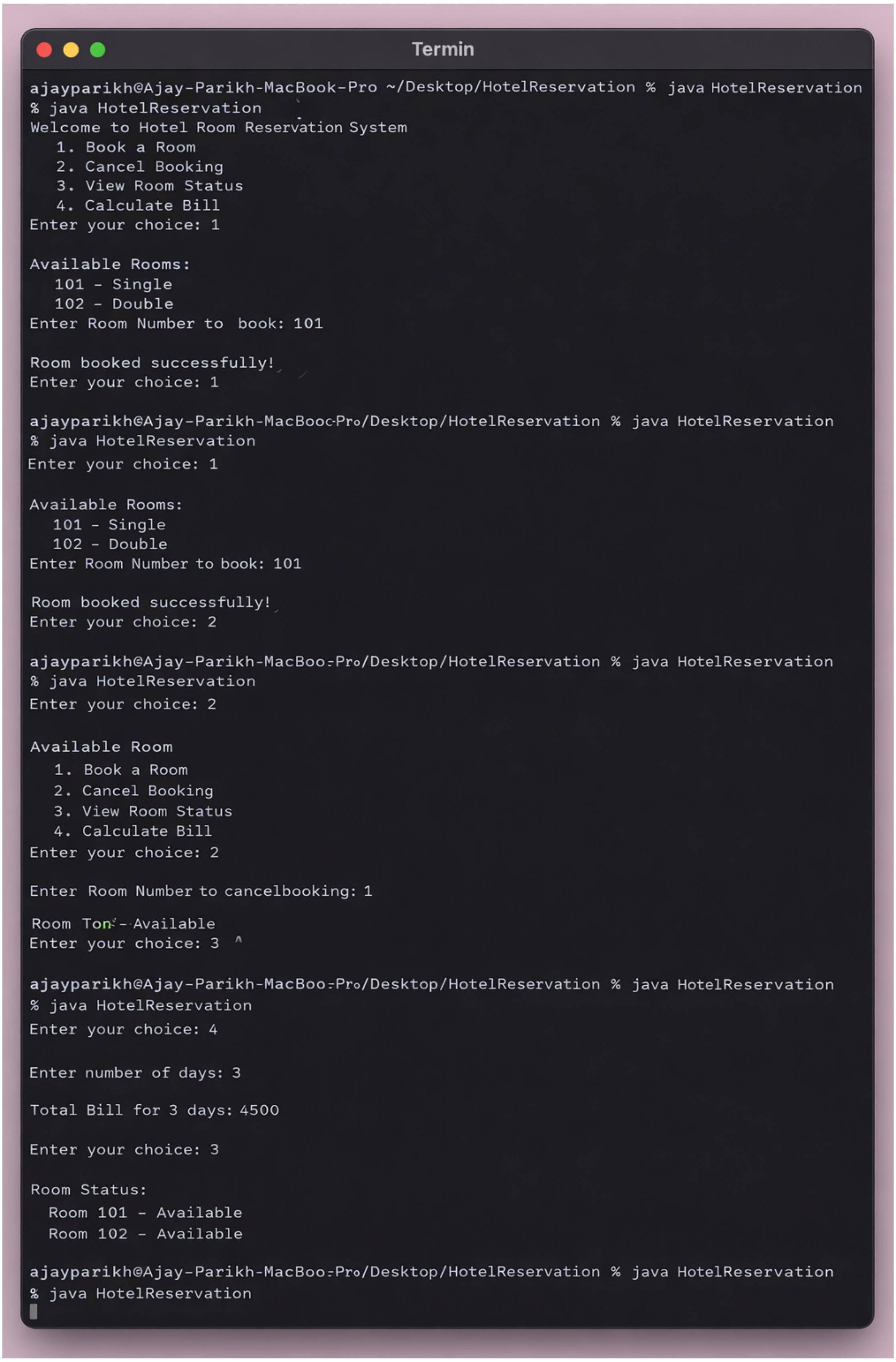
This structured implementation ensures efficient handling of hotel operations and demonstrates the use of abstraction, encapsulation, and modular design.

SAMPLE INPUT / OUTPUT

Room booked successfully
Room already occupied
Booking cancelled successfully

Total Bill for 3 days: 4500

Room Status:
Room 101 – Occupied
Room 102 – Available



Terminal

FUTURE ENHANCEMENTS

- Integration with database for permanent data storage
- Development of a graphical user interface (GUI)
- Online booking and reservation system
- User login and authentication features
- Payment gateway integration for online transactions
- Addition of multiple room types with dynamic pricing

Gantt Chart

A Gantt chart is used to plan and manage the project by dividing it into different tasks such as planning, design, implementation, and testing. It helps in organising the workflow and ensures that each task is completed within a specific time frame.
Even if the project is already designed, proper planning is required for smooth execution and modifications.

The Gantt chart also helps in effective time management by showing when each task starts and when it should be completed. This prevents delays and avoids last-minute work pressure.

1. Planning & Structure

- Identified key features like booking, cancellation, and billing
- Designed system using OOP concepts (classes and interface)
- Divided project into components: Room, Hotel, and Service
- Used 2D array to represent rooms and floors
- Followed a structured approach
- for smooth implementation

2. Time Management

It shows:
- Divided project into tasks (planning, design, implementation, testing)
- Assigned specific time to each task
- Helped in completing work on time
- Reduced last-minute workload
- Improved efficiency and organisation

Conclusion

The Hotel Room Reservation System successfully demonstrates the application of Object-Oriented Programming concepts such as classes, objects, interfaces, and encapsulation. The system provides an efficient way to manage room bookings, cancellations, and billing. It reduces manual effort and improves accuracy in hotel operations.
Overall, the project achieves its objective of simulating a real-world hotel reservation system in a simple and structured manner.

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